

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. Examination (IT)

Dec 2012

IT701 Distributed Computing & Systems

Date: 18.12.2012, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. All questions are compulsory.
2. Indicate clearly, the option you attempt along with its respective question number.
3. Figures to right indicate marks.
4. Rough work is done in the last page of main answer sheet; don't write anything on the question paper.

SECTION – I

- Q - 1 (a) What is InetAddress? Why there is no public constructor in InetAddress class? [05]
- (b) Write the differences between Socket and ServerSocket class. [02]
- Q - 2 (a) What is MulticastSocket? How it differs from DatagramSocket? [04]
- (b) Write a TCP java program that implements a simple client/server application. The client sends the radius of a circle to a server. The server receives the data, uses it to produce the area of the circle as a result and then sends the result back to the client. [07]
- (c) Why java I/O streams can't be used in DatagramSocket? [03]

OR

- Q - 2 (a) List and explain any four Socket exceptions. [04]
- (b) Write a UDP java program in which client will send one dummy packet and server will display the client IP address, port no and content of the packet on screen. [07]
- (c) What are InetSocketAddress and SocketAddress? [03]
- Q - 3 (a) Write an RMI application where client supplies a string and server responds with no. of characters and digits within string. Provide your custom security policy for this application. [06]

OR

- Q - 3 (a) Write an RMI application. Invoke the Remote method having signature `int add(int,int)` and display the result. [06]
- (b) What is stub and skeleton in RMI? What is UnicastRemoteObject? [04]
- (c) "In RMI Objects and primitives are passed by value Remote objects are passed by reference". Justify the statement. [04]

SECTION – II

- Q - 4 (a) What is wsimport utility in web services? Explain the @WebResult Annotation. [04]
(b) List the differences which makes web services differs from other distributed software systems [03]

- Q - 5 (a) Write the differences between parallel computing, distributed computing. [03]
(b) List and explain the characteristics of a distributed system. [04]
(c) What is vector clock? List and explain the basic property of vector clock. [07]

OR

- (c) Explain the Chandy–Lamport algorithm with example. [07]
Q - 6 (a) By taking suitable example explain the consistent global state, strongly consistent, inconsistent state a model of distributed executions. [06]

OR

- Q - 6 (a) What are the algorithmic challenges in distributed computing? [06]
(b) Explain the Flynn's taxonomy of parallel computing with example. [04]
(c) What are the rules to consist while implementing logical clocks? [04]